

ELECTROMAGNETIC LEVITATION AND PROPULSION FOR MODEL TRAINS

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FINAL REPORT

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A handwritten signature in black ink, reading "Justin Smallwood". The signature is written in a cursive, flowing style with a large initial 'J' and a long, sweeping underline.

Abstract

As transportation continues to develop and modernize, the evaluation of alternative and innovative systems will be essential to meet future mobility needs. Magnetic levitation (maglev) has been identified as a promising system for transportation, and its technology has been widely implemented in trains; applications of electromagnetic propulsion systems have been implemented in high-speed trains across Europe and Asia. This thesis provides a methodology for designing and building a magnetic train model. Using microcontrollers, transistors, and Time-of-Flight distance sensors, this model investigates how permanent magnets and electromagnetic coils can be coordinated to control the movement of a levitating train.

Two design iterations were evaluated to achieve stable levitation and propulsion. The first utilized coils wound with 30 AWG magnet wire mounted on electromagnetic supports to stabilize the train along the track. Lateral stabilization of the train requires sufficient support from the bottom permanent magnets. This gauge wire emitted too much heat and was unable to provide sufficient force on the train to achieve stable levitation. A second iteration using 20 AWG magnet wire eliminated heat-related failures but was unable to produce enough force to achieve a stabilized system.

Although a fully levitating prototype was not realized, the sensor and control system performed as intended; the Arduino reads the train's positional data and updates the electromagnet duty cycles in real time. These results highlight the primary limitations of electromagnets and validate the control approach. This report details the development of the train and track using CAD software, the design considerations involved in the construction of the physical and electrical systems of the model, and future work to achieve stable levitation.

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List of Symbols

F	Force in Newtons (N)	4
g	Air gap; distance between magnet and point in meters	4
μ_0	Magnetic permeability of free space (henries per meter)	4
A	Cross-sectional area of the magnet in m^2	4
N	Number of turns in the coil	4
I	Current in amperes	4
μ_r	Relative permeability of a material	22

Chapter 1

Introduction

1.1 Background

Electromagnetic propulsion may emerge as a promising technology in the future of transportation. The technology has been popularized largely by the development and promotion of magnetic levitation (maglev) trains. Maglev trains offer significant societal benefits through faster travel, sustainable infrastructure, and opportunities for economic expansion [7]. These trains can exceed 310 mph due to the absence of rail friction [1]. The low friction environment is maintained by minimizing contact between the train and the track. Early conceptualizations of magnetic levitation technology were proposed by American physicist Robert H. Goddard; the idea was later refined in the 1960s [5].

There are two main classifications of electromagnetic suspension: electromagnetic suspension (EMS) uses attractive forces, and electrodynamic suspension (EDS) uses repulsive forces [2]. In the EMS system, the “bottom of the train wraps around a steel guideway,” leaving approximately 1 centimeter between the train and the track [1]. In the EDS system, the train levitates nearly 4 inches above a U-shaped guideway [1]. The EDS system also requires wheels to assist with liftoff during propulsion [1]. Both the EMS and EDS trains switch magnetic poles on the guideways to propel the train [1].

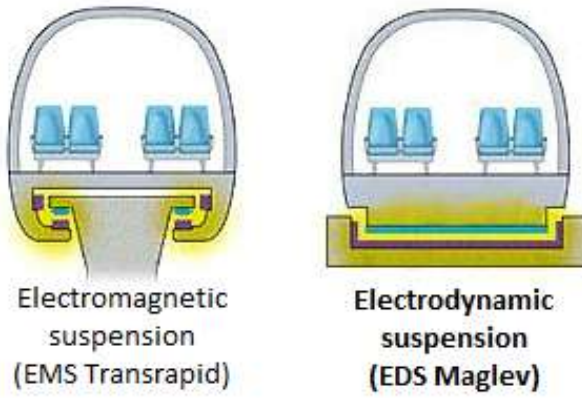


Figure 1.1: EMS vs. EDS.

1.2 Motivation

A major inspiration for the design of this thesis is from the YouTube video “Building a Maglev train for my Christmas Tree” by the creator Hyperspace Pirate, in which permanent magnets and evenly spaced solenoids are used to propel a 3D-printed train around a circular track [9]. While this YouTube project was designed to allow the train to remain in contact with the track, this thesis focuses on the development of a maglev train that operates without physical contact. The design requirement for this thesis is that the system is capable of levitating and propelling the model train.

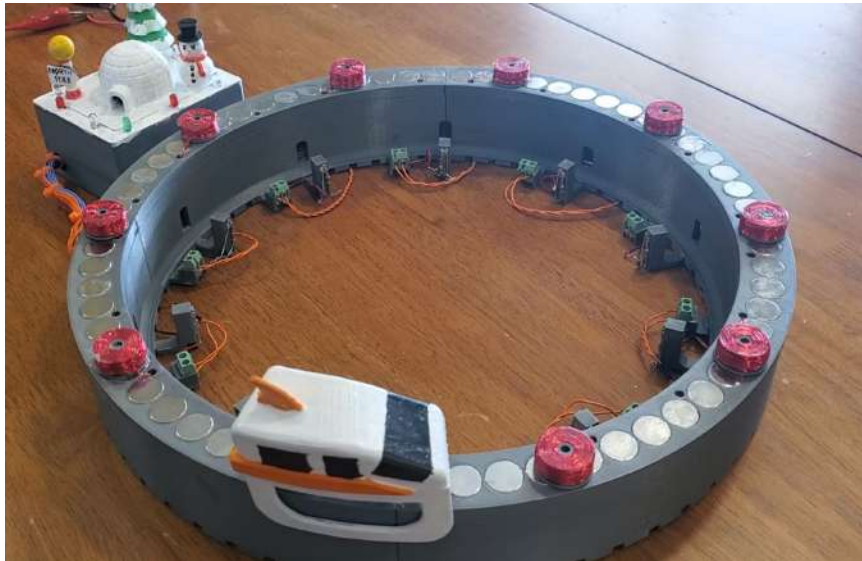


Figure 1.2: Hyperspace Pirate maglev train model [9].

Chapter 2

Methods of Propulsion

2.1 Propulsion

This thesis is largely dependent on the train’s ability to achieve effective propulsion across the track. By utilizing a combination of permanent magnets and electromagnets, the maglev model train can be propelled in several configurations. Permanent magnets are mounted on the bottom of the train to remove the need for an onboard power source, reducing the train’s overall weight. According to Earnshaw’s theorem, “it is impossible to place a collection of bodies, subject only to electrostatic forces, in such a way that they remain in a stable equilibrium configuration” [11]. Hence, electromagnets are used in this project to stabilize the system.

Two propulsion arrangements are considered: one configuration places electromagnets beneath the track and permanent magnets along the guide rails, while the other places permanent magnets beneath the track and electromagnets on the guide rails. The latter configuration was selected to provide greater control over the train’s motion along the track.

Magnetic levitation trains are propelled across a railway using a linear motor that creates alternating magnetic fields, which push and pull the train along the track. In the Hyperspace Pirate’s maglev model, propulsion is achieved by using permanent magnets on the bottom of the track for levitation and brief bursts of force from electromagnets to push the train to the subsequent electromagnet. Although this model is an EDS maglev system, this creator uses overhanging sides on the train to stabilize it, removing the need for additional horizontal stability.

In this thesis, propulsion is implemented using horizontal electromagnets and a

direct current power supply. Due to its inability to alternate magnetic fields, this model cannot fully replicate a conventional EDS system. Thus, this thesis investigates the feasibility of modeling an EDS maglev train using direct current for horizontal propulsion.

2.2 Vertical Electromagnetic Propulsion

The target vertical force of the electromagnet was established by acquiring the weight of the model train in Newtons and applying an appropriate safety factor of 1.7 to account for potential errors or variations in the magnetic field. Equation 2.2.1 is then used to calculate the number of turns needed to achieve the target force for several given current levels.

$$NI = \sqrt{\frac{2Fg^2}{\mu_0 A}} \quad (2.2.1)$$

The current and number of turns are used to determine the voltage required for the chosen magnetic wire to create a field strong enough to lift the model train. Based on the train's weight of 152 grams, the amount of downward force produced by the train was calculated in Newtons, with the added safety factor. The air gap was estimated to be 0.4 inches, the diameter of the coil was 1 inch, which means that the cross-sectional area was approximately $5 \times 10^{-4} m^2$. The resulting current levels, voltages, and power based on these variables are presented in Tables 2.1 and 2.2. There is no variation in the number of turns between the 20 AWG and 30 AWG coils; instead, the coil configuration remains fixed and the required voltage is adjusted to account for the difference in wire gauge.

Current I	No. of Turns N	Total Resistance (Ω)	Req. Voltage (V)	Power (Watts)
2.5	362.63	0.96	2.41	6.02
3	302.19	0.80	2.41	7.23
4	226.64	0.60	2.41	9.64
5	151.10	0.48	2.41	12.05

Table 2.1: Resistance, voltage, and power for vertical electromagnet levitation (20 AWG).

Current I	No. of Turns N	Total Resistance (Ω)	Req. Voltage (V)	Power (Watts)
2.5	362.63	9.80	24.49	61.23
3	302.19	8.16	24.49	73.48
4	226.64	6.12	24.49	97.97
5	151.10	4.90	24.49	122.47

Table 2.2: Resistance, voltage, and power for vertical electromagnet levitation (30 AWG).

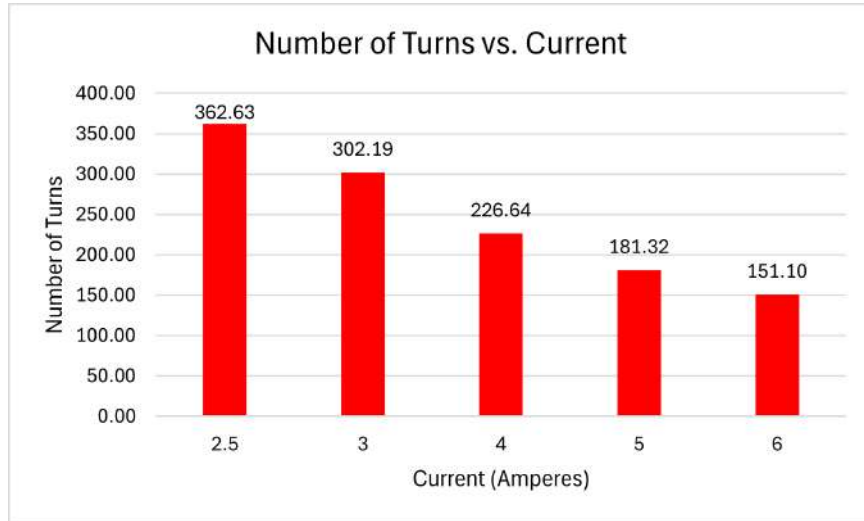


Figure 2.1: Number of Turns vs. Current for vertical electromagnet levitation producing 2.5 N (20 AWG & 30 AWG).

2.3 Horizontal Electromagnetic Propulsion

For the propulsion arrangement with side-mounted electromagnets, the target horizontal force of the electromagnet was estimated using the force values proposed for the vertical electromagnetic propulsion system in the initial design. In this model, the safety factor of the horizontal propulsion system was increased to 3 to compensate for the reduction in magnetic field strength caused by variations in the duty cycle.

Current I	No. of Turns N	Total Resistance (Ω)	Req. Voltage (V)	Power (Watts)
2.5	301.08	0.80	2.00	5.00
3	250.90	0.67	2.00	6.00
4	188.18	0.50	2.00	8.00
5	150.54	0.40	2.00	10.00

Table 2.3: Resistance, voltage, and power for horizontal electromagnet stabilization (20 AWG).

Current I	No. of Turns N	Total Resistance (Ω)	Req. Voltage (V)	Power (Watts)
2.5	301.08	8.13	20.34	50.84
3	250.90	6.78	20.34	61.01
4	188.18	5.08	20.34	81.34
5	150.54	4.07	20.34	101.68

Table 2.4: Resistance, voltage, and power for horizontal electromagnet stabilization (30 AWG).

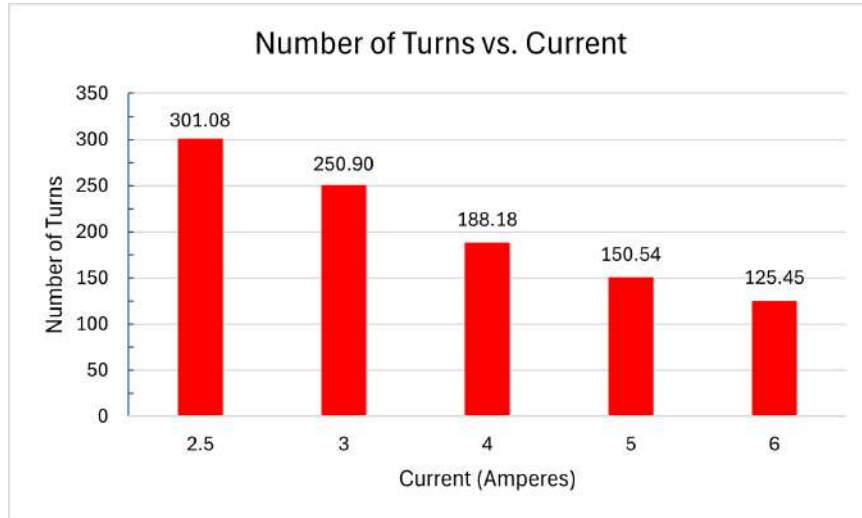


Figure 2.2: Number of Turns vs. Current for horizontal electromagnet stabilization producing 4.5 N (20 AWG & 30 AWG).

In the final design iteration, the horizontal electromagnetic field strength was estimated after maxing out the amperage of the DC power supply and creating the

largest sized coil possible that the electromagnetic support could contain; thus, 50 turns of 20 AWG magnet wire were used.

No. of Turns N	I	Target NI	Force (N)	Total Resistance (Ω)	Req. Voltage (V)
50	4	200	3.16×10^{-1}	1.33×10^{-1}	5.31×10^{-1}

Table 2.5: Resistance, voltage, and power for horizontal electromagnet stabilization in the final design (20 AWG).

Chapter 3

Engineering Design

3.1 Introduction

The goal of this thesis was to design a maglev model train that makes no contact with guide rails or the ground, with the train and track designed using PTC Creo. Additionally, the circuit and code were designed to enable autonomous movement of the train around the track. Establishing stable magnetic control of the train and optimizing motion smoothness are beyond the scope of this project. Controlling the train's speed or height is also outside the scope of this thesis.

Requirements	Constraints
Vertical Levitation	The train must not contact the track bottom or exceed the height of the guide rails.
Horizontal Stability	The train must fit inside the straight and curved segments of the track, with adequate side clearance for movement.
Scalability	The code must be scalable to support the addition of more track segments.

Table 3.1: Design requirements and constraints.

The objective evaluation criteria for the thesis are the following:

- The code must be able to recognize the train's position and movement. It must also be expandable to accommodate additional track segments.
- For levitation, the permanent magnets must generate sufficient repulsive force to counteract the weight of the train.
- The electromagnets must be capable of quickly producing a magnetic field that can repel a permanent magnet. Additionally, the electromagnets must be strong enough to stabilize the train in the center of the track.

3.2 Final Design Presentation

The final design consisted of three ToF sensors and six MOSFET modules, with each module driving two electromagnets. Flyback diodes were connected across each MOSFET module to protect the circuit from high voltage spikes generated when the electromagnets were switched off. The sets of electromagnets were evenly spaced across two straight sections of the track. The arrangement of the electromagnets on the track is shown in Figure 3.3.

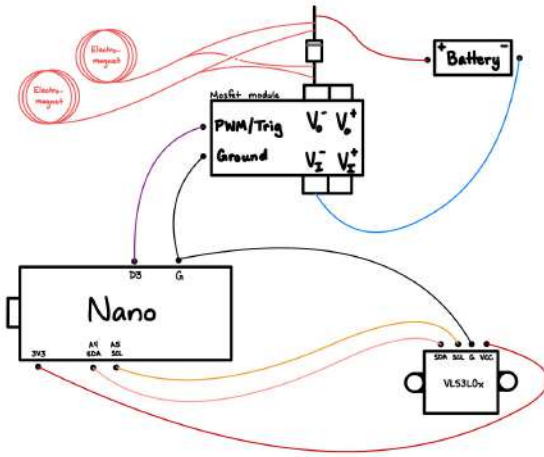


Figure 3.1: Drawn image of subsystem of electromagnets.

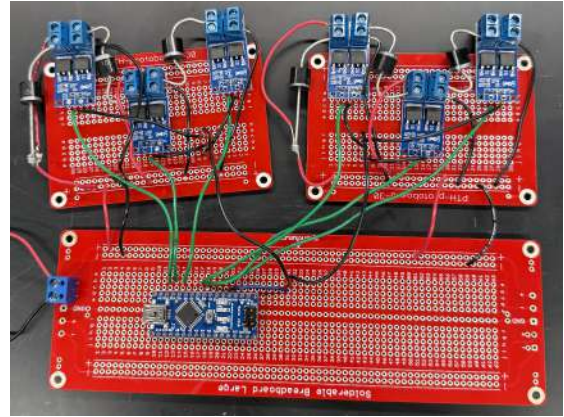


Figure 3.2: Image of final design setup on the track.



Figure 3.3: Front view of electromagnet circuit final assembly.

The code for this system was adapted for use with 20 AWG wire. Because the power supply provided direct current, the propulsion of the train was achieved by varying the strength of the magnetic fields generated by the electromagnets. Duty cycles of 50%, 90%, and 100% were implemented to produce different magnetic field strengths, depending on which sensor detected a nearby object. The two sets of electromagnets surrounding the activated sensor were driven at a duty cycle of 90%, while the subsequent sets were driven at 50%. When the ‘R’ or ‘L’ key was pressed on the keyboard, the train was propelled in the corresponding direction by increasing the duty cycle of the opposite set of electromagnets to 100%. This code is outlined in Appendix B.

The engineering standards used in the design are listed below:

Requirements	Description
Track Clearances	There is a 0.01 inch nominal clearance in the interlocking track joints to accommodate 3D printing variations.
Electromagnet Support Clearances	There is a nominal clearance of 0.005 inches provided on the interior of the electromagnet supports to ensure a tight fit with the low-carbon steel rod.

Table 3.2: Engineering standards.

3.3 Evaluation of the Design

3.3.1 Procedure

To construct an electromagnet with a DC power supply, magnet wire is first wound into a dense coil. After determining the desired magnetic field direction, the two ends of the wire are connected in series with a flyback diode. The flyback diode should be oriented opposite to the intended current flow through the electromagnet. Finally, the DC power supply is connected to the circuit. This system allows current to flow through the electromagnet while preventing component damage from an inductive kickback, which is a “very rapid change in voltage across an inductor” when the power supply is switched off [4].

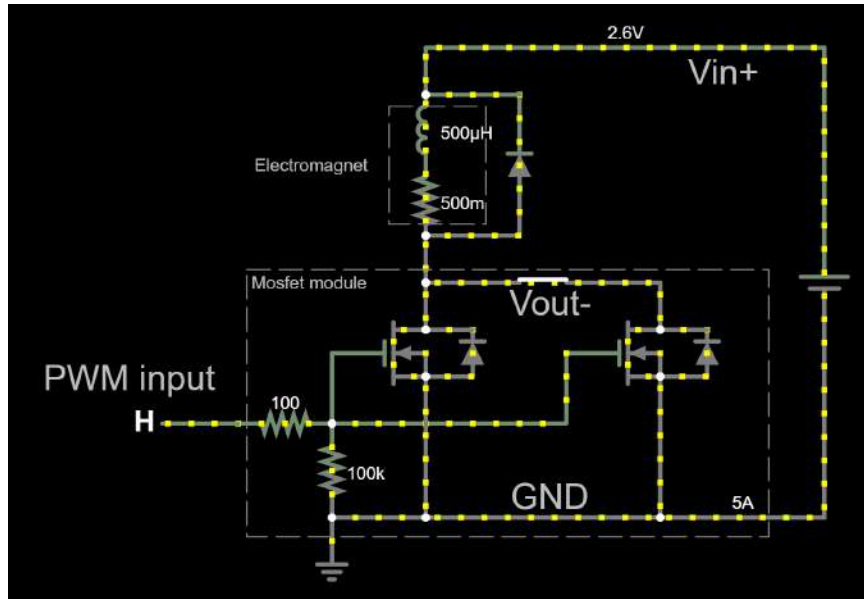


Figure 3.4: Electrical circuit schematic for a single electromagnet and MOSFET module.

The strength of an electromagnet is primarily determined by the current flowing through the wire; increasing the current increases the magnetic field strength. The voltage of the DC power supply can be adjusted as needed, provided that the current remains the limiting factor in the system. However, the current is ultimately constrained by the power supply’s maximum current rating and heat generation in the magnet wire.

Accordingly, design requirements were established to determine the appropriate magnet wire and the model’s ability to levitate the train using permanent magnets

and stabilize it using electromagnets. The main design requirements for this thesis are as follows:

- RQ1: Low heat generation
- RQ2: Stable permanent magnet configuration
- RQ3: Stable side electromagnets
- RQ4: Individual control of the track segment electromagnets
- RQ5: VL53L0X sensors used to detect the train's position
- RQ6: Scalable circuit design
- RQ7: Scalable train and track design

Design 1: Full track levitation (30 AWG)

Design 2: Track segment electromagnet control (20 AWG)

	Eval. of RQ1	Eval. of RQ2	Eval. of RQ3	Eval. of RQ4	Eval. of RQ5	Eval. of RQ6	Eval. of RQ7
Design 1	X	X	X	✓	✓	✓	✓
Design 2	✓	X	X	✓	✓	✓	✓

Table 3.3: Iterative process.
(✓ - Standards met, X - Standards not met)

3.3.2 Design of the Train and Track

The model track is designed to replicate the EDS system illustrated in Figure 1.1, with slight modifications. The track has a U-shaped configuration, with straight segments measuring 8 inches and curved sections having an inner radius of 7 inches. Permanent magnets are mounted along the bottom of the track, while electromagnets are positioned along the sides for stability and propulsion. The track walls extend 2 inches above the base, providing mounting support for both the electromagnets and multiple VL53L0X Time-of-Flight (ToF) sensors that monitor the train's motion along the track.

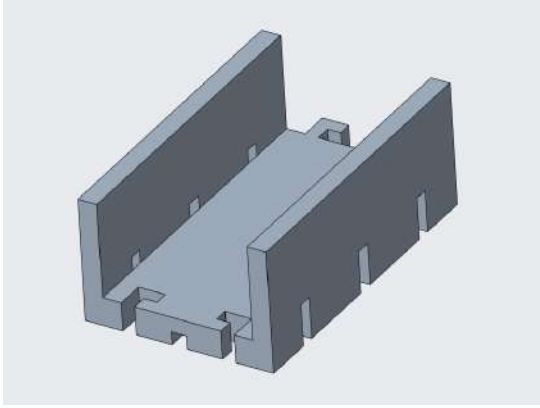


Figure 3.5: Straight track.

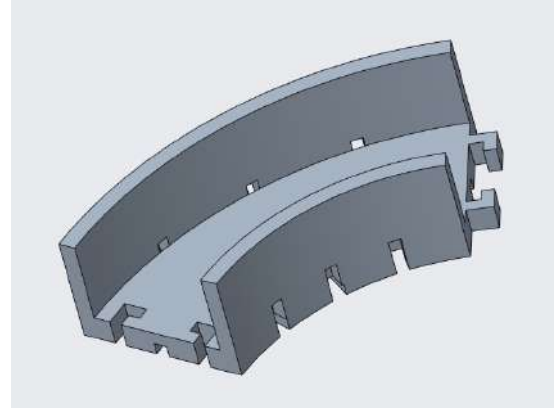


Figure 3.6: Curved track.

An electromagnetic support device was developed to hold the coils in a solenoid shape. The support features a hollow core capable of securing a 1-inch iron core and is also designed to hook over the guide rails of the track for stability.

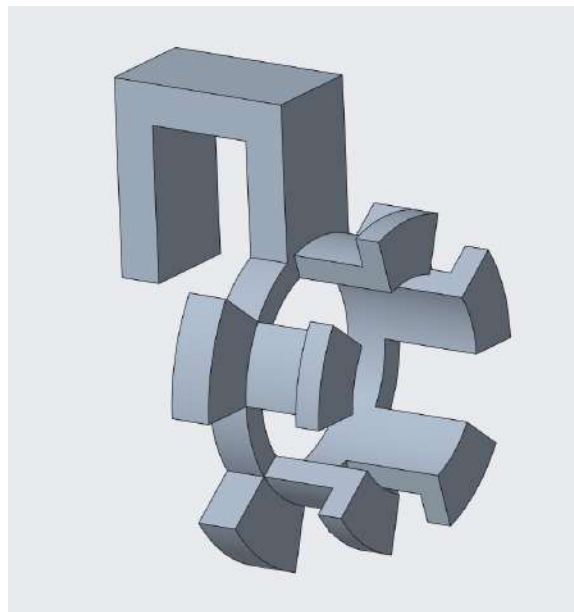


Figure 3.7: Electromagnet support.

The model train is 6 in. x 2 in. x 1.5 in. Its front features a curved nose inspired by the Japanese SCMaglev design, which reduces aerodynamic drag and improves efficiency. The rear of the train includes a hook that allows additional trains to be coupled to the lead unit. The inner diameter of the U-shaped track is 4 inches, providing sufficient tolerance for variations in the train's motion. This clearance also allows the train to navigate the track curves smoothly without contacting the guide rails.

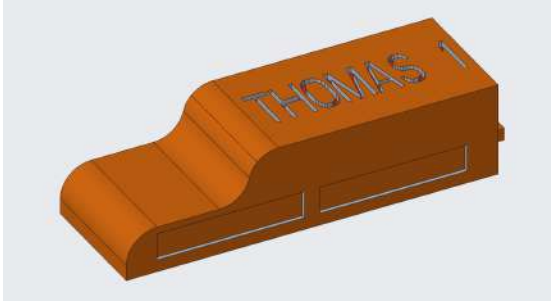


Figure 3.8: Model train.



Figure 3.9: Japanese SCMaglev [3].

Appendix B includes MATLAB code that calculates the minimum clearance required for the train to traverse track curves without contacting the guide rails. The track is divided into 8-inch segments, which are connected using L-shaped joints at both ends to ensure stability and alignment.

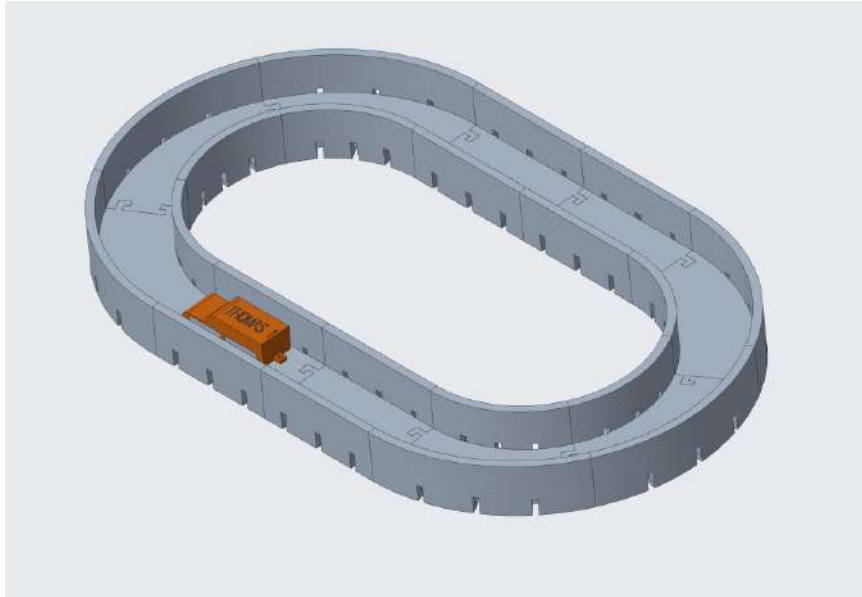


Figure 3.10: Model track.

Grooves are incorporated beneath each track segment to accommodate wiring. Holes are also included at the midpoint of each segment on both sides, allowing magnet wires to be routed along the guide rails. This design improves the overall organization and manageability of the system wiring.

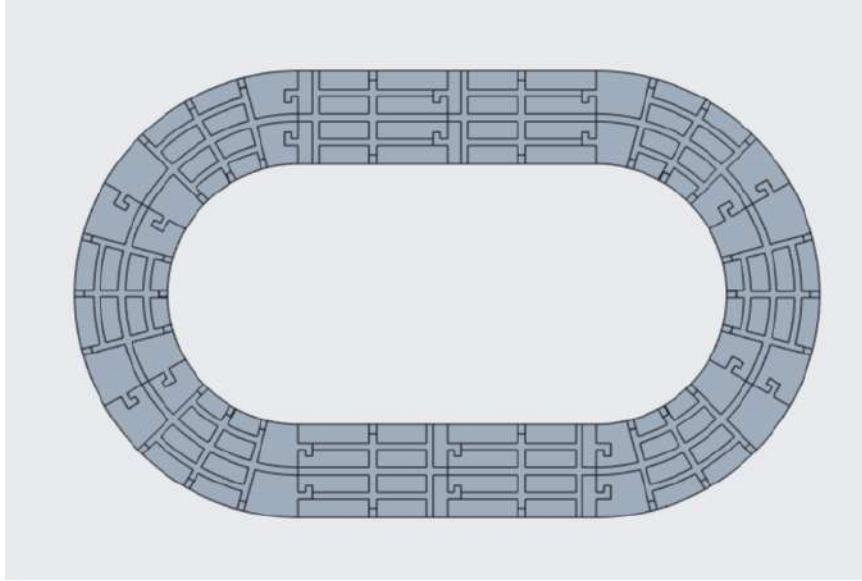


Figure 3.11: Full track bottom view.

The interlocking joints of the tracks were designed with a nominal clearance of 0.01 inches to account for variations in the 3D printing while maintaining a secure fit.

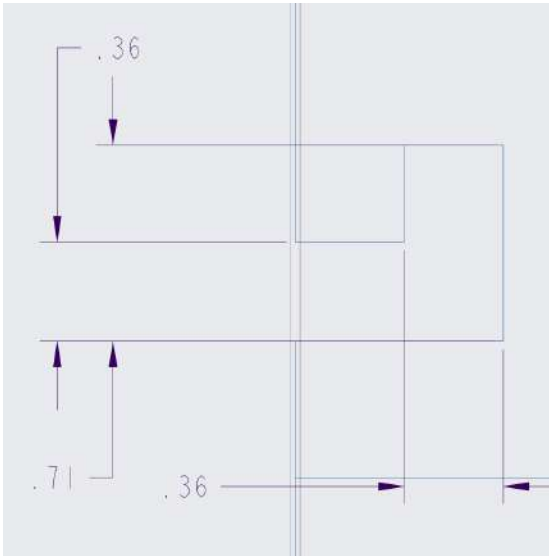


Figure 3.12: Engineering drawing of inner track joint.

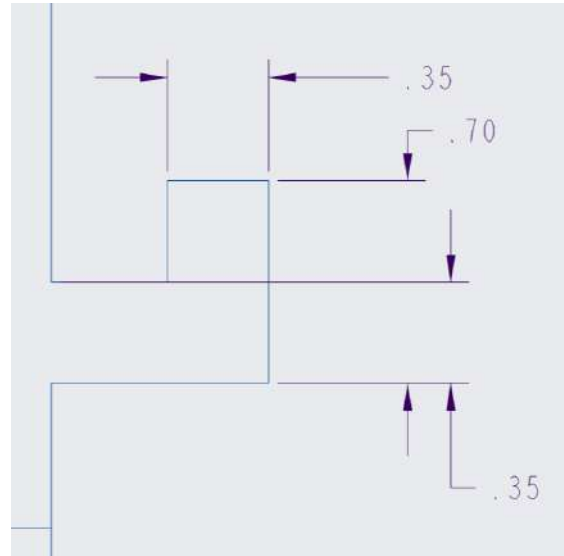


Figure 3.13: Engineering drawing of outer track joint.

3.3.3 Circuits and Microcontrollers

The circuit design for this thesis was modeled after the Hyperspace Pirate’s maglev model. Modifications were made to accommodate the use of different components and a custom track design.

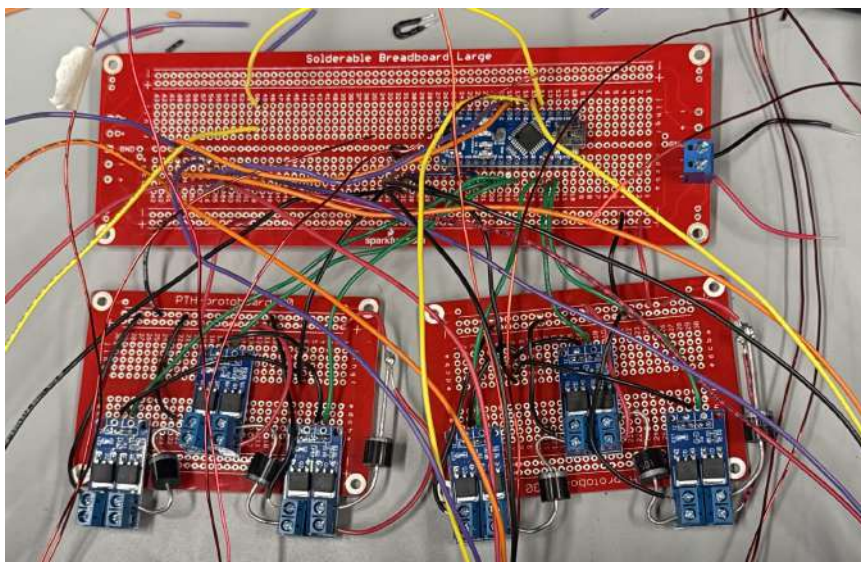


Figure 3.14: Electromagnet circuit.

During the development of the Arduino code, two propulsion sequencing methods were considered: activating electromagnets based on predetermined timing within specific track segments or dynamically controlling the magnetic fields using real-time ToF sensor data. The initial design was based on the latter method. In the first design, an Arduino Nano controlled multiple Seeed Xiao RP2040 microcontrollers, which in turn controlled the VL53L0X ToF sensors and MOSFET modules. The RP2040 microcontrollers managed the activation sequence of electromagnets for each track segment as the train approached, while the Arduino Nano organized their overall operation. The MOSFET modules allowed for control over the duty cycles of the electromagnets, and the ToF sensors determined the train’s position, triggering the corresponding RP2040 to initiate the appropriate electromagnetic sequence.

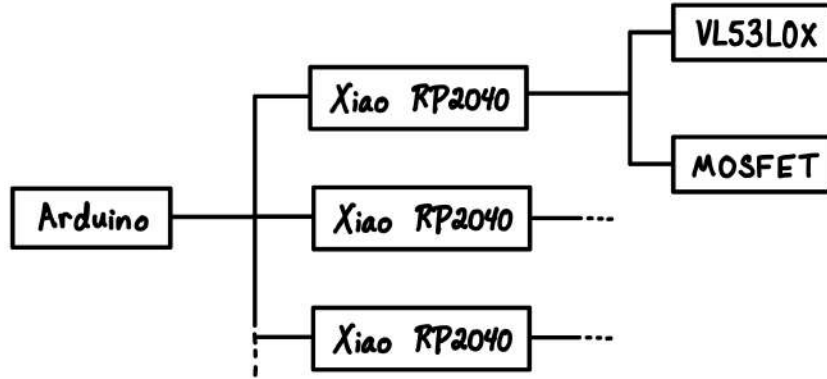


Figure 3.15: Original system architecture.

This approach was abandoned because, although the ToF sensors operate at 3.3 V, their output signals can exhibit instability and may exceed voltage levels acceptable to the RP2040. As a result, the Arduino Nano was used to read the ToF sensor data directly. With multiple sensors connected to the Nano, the device’s I2C address for each ToF sensor had to be reassigned using the XSHUT pins and code.

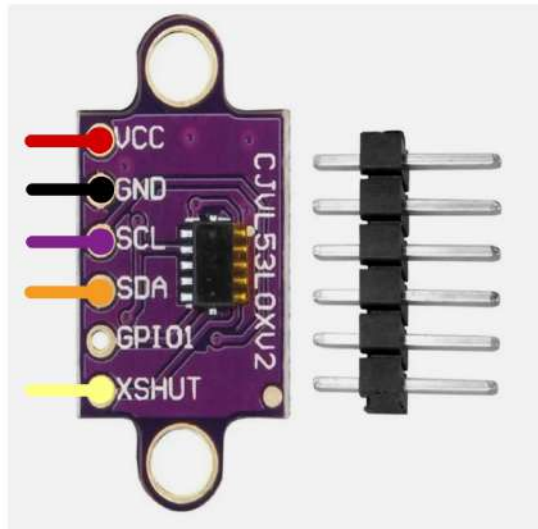


Figure 3.16: XSHUT pin and ToF sensor [10].

3.3.4 Permanent Magnets

The size of the permanent magnets plays a critical role in their ability to levitate the model train. Two types of magnets were tested: smaller magnets with a diameter of 0.25 in., grade N35, and a maximum pull force of 1.2 lbf; and larger magnets with a diameter of 1.25 in., grade N30, and a pull force of 18 lbf. In the first iterations of the

design, smaller neodymium magnets were tested in several configurations, including aligning the poles in the same orientation and arranging them in a Halbach array as shown in Figure A.1. The main obstacle with these magnets was their relatively localized magnetic field, which resulted in a limited effective levitation range and instability in maintaining the train's position. None of these orientations seemed reasonable, and the variability in experimentation was further limited by the practicality and cost of using smaller magnets.

In contrast, the larger magnets performed significantly better, as their increased surface area generated broader and more spatially extensive magnetic fields. A limitation of the larger magnets is that they cannot be easily oriented into a Halbach array to further enhance the magnetic field strength. Despite this limitation, the magnetic force produced by the larger permanent magnets was sufficient to counteract the weight of the train.

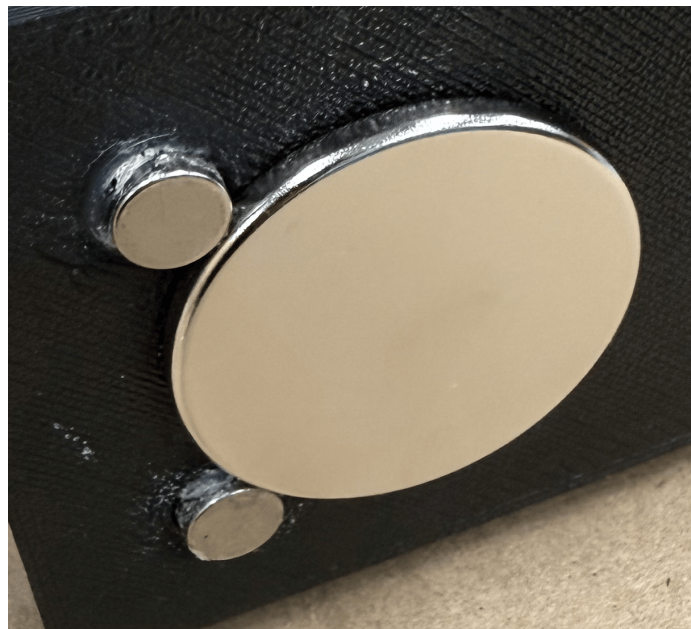


Figure 3.17: 1.25 in. neodymium magnet next to two 0.25 in. neodymium magnets.

Two main configurations were tested with the larger magnets: pairs of two and an X-shaped configuration of five magnets. These arrangements were selected to improve stability, as the train itself used only two onboard magnets. The paired magnet configuration produced a strong magnetic field when the train's magnets were positioned directly above the junction between the two track magnets. However, as the train moved beyond this point, there was a noticeable reduction in magnetic field strength caused by gaps in field coverage; this region is shown in Figure 3.18.

When the paired magnet configuration is arranged in series, this effect becomes more pronounced, as the train dips between magnet pairs and only regains its original height when centered over the next pair.



Figure 3.18: Series of paired magnets.

The other configuration tested with the larger magnets was the X-shape arrangement. This orientation not only used the fewest magnets overall but also minimized the total gap area, reducing regions of weak magnetic field strength. Although the X-shape configuration contained a relatively large non-magnetized region, the electromagnetic fields were intended to align and stabilize the train along the track centerline, eliminating the need for magnetic coverage across the full track width.

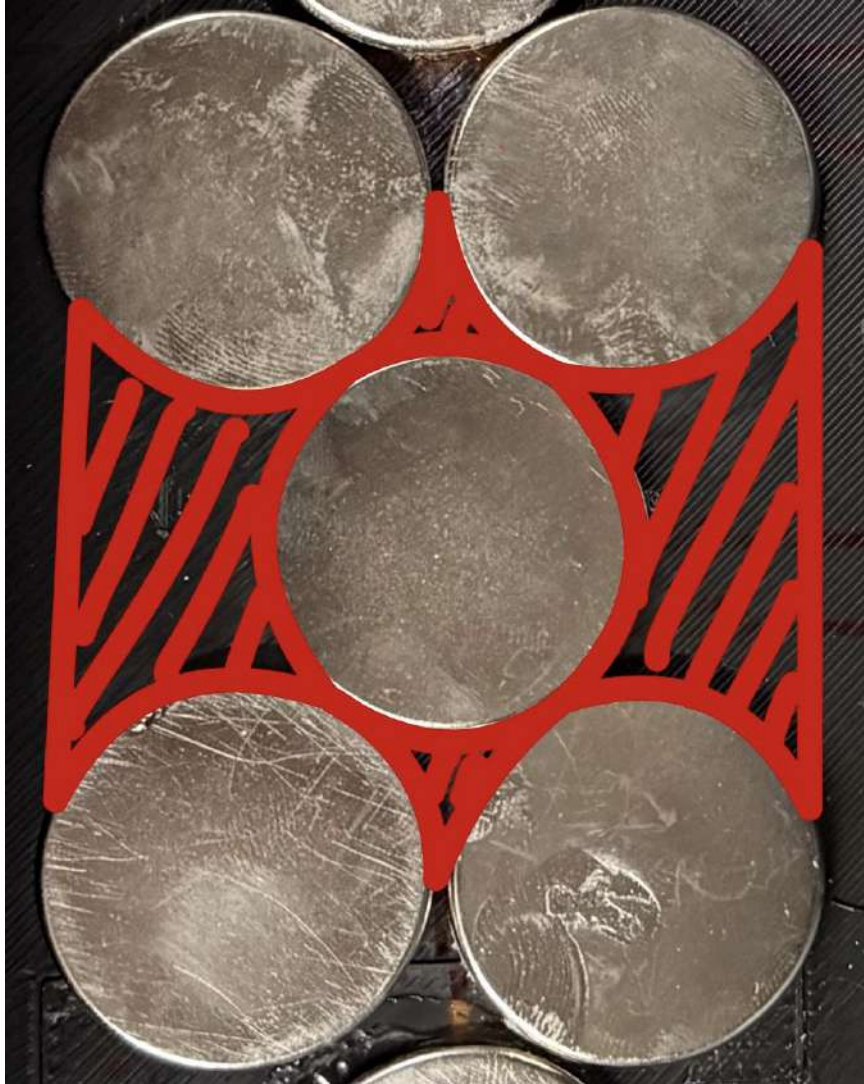


Figure 3.19: “X” magnet orientation.

3.3.5 Electromagnets

Magnet Wire

Initially, 20-gauge magnet wire was selected for generating the magnetic field to stabilize the train. However, 30-gauge magnet wire was tested in the first iteration of the project so that more coils could be packed into a solenoid shape as illustrated in Figure 3.20. Although this reduced the allowable current, the smaller coil diameter enabled a denser coil, increasing the overall magnetic field strength; a thinner wire gauge was also used for propulsion in Hyperspace Pirate’s maglev model.

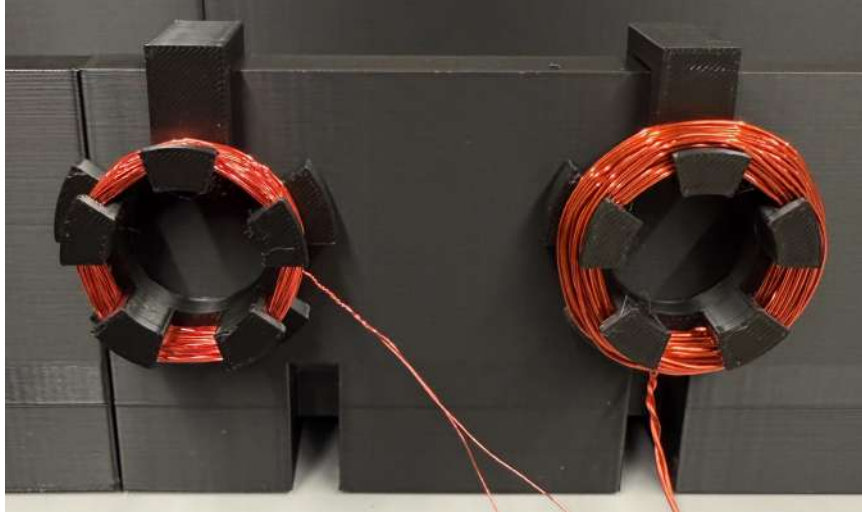


Figure 3.20: 20 AWG next to 30 AWG.

The major concern for using a thinner wire gauge was heat generation. As shown in Table 2.4, the increased coil density combined with the high resistance of the wire resulted in substantial heat production with the 30 AWG wire. Due to limited cost-efficient options for improving airflow and cooling within the track, tests with reduced duty-cycles were conducted to prevent the electromagnet from reaching excessive temperatures. A duty cycle reduction to 35% optimally reduced the electromagnet's on-time while maintaining sufficient magnetic field strength. However, this reduction did not meet the design requirements for slower, more controllable magnetic propulsion.

Ultimately, the 20 AWG wire reduced the need for a low duty cycle, as it did not generate significant heat at 5 A, which was the maximum current supplied by the DC power supply. Despite space constraints limiting the number of turns in the electromagnetic support, a comparable magnetic field strength was achieved.

The available quantities of 20 AWG and 30 AWG magnet wire were also a large consideration in the design of the project. The wire was purchased by total weight. Because 20 AWG wire is thicker and weighs more, there was a limited length of 20 AWG wire compared to the greater availability of 30 AWG wire. As a result, even when winding 150-turn coils on the electromagnetic support with 30 AWG magnet wire, significantly more coil sets could be produced compared to 20 AWG wire.

Design Criteria	20 AWG	30 AWG
High coil density	X	✓
High coil length per 4 oz.	X	✓
Minimal heat production	✓	X
Magnetic force sufficient for stabilization	✓	✓

Table 3.4: Iterative design of 20 AWG and 30 AWG magnet wire.
(✓ = Standards met, X = Standards not met)

Core

Cores are commonly used in electromagnets because they increase the magnetic field strength by a factor related to the material’s relative permeability, μ_r [8]. Iron is typically the preferred core material due to its high relative permeability. In this project, low-carbon steel, which had been procured in place of pure iron, was tested, as it consists of approximately 99% iron. Although low-carbon steel has a chemical composition similar to iron, it remains ferromagnetic and is therefore classified as a magnetic material [8].

Nonetheless, the low-carbon steel did not multiply the magnetic field strength of the 20 AWG or 30 AWG wire. Instead, it resulted in an unintended attraction between the electromagnet and the train, even when like poles were oriented to repel. The electromagnet field strength was theorized to be not strong enough to overcome the strength of the low-carbon steel, making it an unsuitable material for this application.

Chapter 4

Results

4.1 Full Track Maglev Model

During testing with 30 AWG magnet wire, it quickly became evident that this gauge was not suitable for horizontal electromagnetic propulsion. After approximately 2 seconds of operation using the duty-cycle evaluation code, the wire began to emit smoke due to excessive resistive heating. The heat also caused deformation of the PLA electromagnetic support. Furthermore, reduced duty cycles did not generate a magnetic field strong enough to stabilize the model train at the center of the track.



Figure 4.1: Full track maglev model.

4.2 Electromagnetic Control Model

The 20 AWG magnet wire showed no signs of thermal failure when operated at a 100% duty cycle. The thicker wire did reduce the total length of wire that could be wound onto the electromagnetic support; however, this did not have a significant impact on the system performance. In the end, the electromagnets did not achieve the expected field strength and failed to stabilize the train; the train would immediately fall into the corners of the straight track as shown in Figure 4.2. The permanent magnets also failed to levitate the train. This outcome was expected, as the vertical stability of the system depended on the horizontal electromagnets maintaining the train's position at the center of the track.

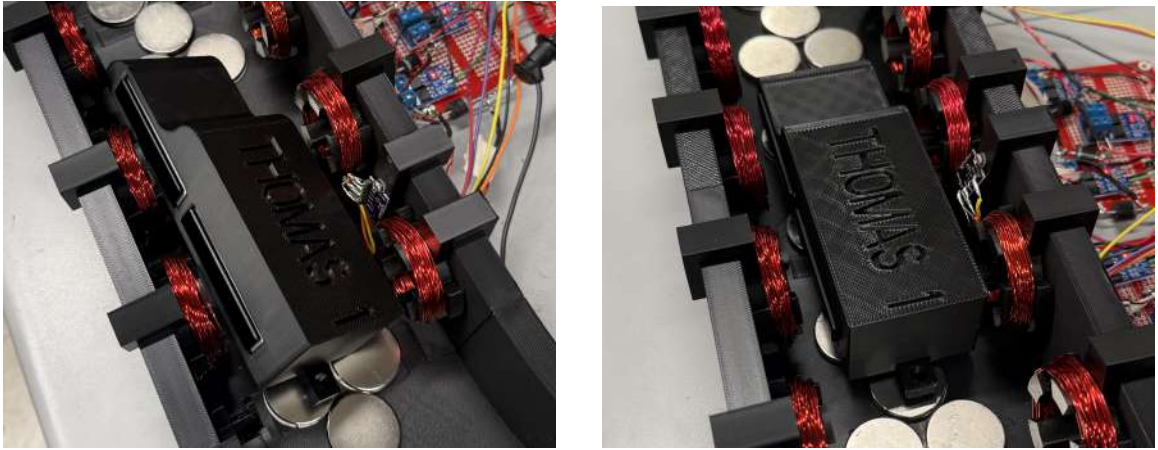


Figure 4.2: Train failure.

With the full range of duty cycles available, the code was designed to control the movement of the train across the track. The code was successful in quickly acquiring positional changes in the train's movement and updating the surrounding magnetic fields appropriately. The code could not be refined since the train was unable to levitate and stabilize.

Chapter 5

Conclusion and Discussion

5.1 Objectives Met

While this project did not succeed in producing a fully functional model, it yielded valuable insights into the selection of materials for developing an electrodynamic suspension (EDS) maglev model train system. In addition, the thesis demonstrated the integration of MOSFET modules and VL53L0X Time-of-Flight sensors, which showed potential for precisely controlling the position of a maglev model train.

5.2 Conclusion

The physical design of the train and track system played a crucial role in enabling the intended wiring layout and supporting the project's magnetic design objectives. The model's inability to stabilize is likely due to regions of reduced magnetic field strength from the permanent magnets, in addition to the insufficient strength of the electromagnets.

Improving the strength of the electromagnets would likely involve finding a DC power supply that can support amperage higher than 5 A or designing a larger electromagnet support that could support more than 50 turns of 20 AWG magnet wire. Additionally, more turns of magnet wire for this gauge would likely require widening the track, as there is limited clearance between the supports and the train. With stronger horizontal magnetic forces and improved electromagnetic control, there is potential for the system to achieve stable levitation and sustained operation of the model train.

5.3 Future Work

In continuation of this model’s current experimentation, several possible approaches could be investigated. One option is to reduce the width of the track, bringing the train closer to the electromagnet and increasing its interaction with the horizontal magnetic fields, potentially improving stability. Another possible improvement is replacing the circular permanent track magnets with square magnets to minimize dead spots in the magnetic field. Additionally, incorporating a soft iron core could be explored to increase the strength of the electromagnets.

Future iterations of this project could also explore the use of electromagnets mounted on the floor of the track. Hall effect sensors could be integrated alongside these electromagnets to improve control of the train’s positioning by enabling real-time adjustments to the magnetic field for stabilization. Another extension would be to use electromagnets for both levitation and propulsion of the train.

An alternative direction would be to design an EMS maglev model. However, constructing an electromagnetic suspension (EMS) system at a model scale would be challenging, as the reduced tolerances significantly increase the likelihood of contact between the train and the track. It would also require embedding sensors within the track to accurately determine the location of the train and monitor its motion.

Finally, further investigation could be conducted into how electromagnet geometry influences levitation and propulsion of the train, potentially drawing inspiration from designs such as figure-eight-shaped superconducting magnet coils. The use of an AC power supply could also be explored as a means of achieving a more conventional maglev operating system.

Chapter 6

Engineering Design Tables

6.1 Introduction Table

Requirements	Constraints
Vertical Levitation	The train must not contact the track bottom or exceed the height of the guide rails.
Horizontal Stability	The train must fit inside the straight and curved segments of the track, with adequate side clearance for movement.
Scalability	The code must be scalable to support the addition of more track segments.

Table 6.1: Design requirements and constraints.

6.2 Design Presentation Table

Requirements	Description
Track Clearances	There is a 0.01 inch nominal clearance in the interlocking track joints to accommodate 3D printing variations.
Electromagnet Support Clearances	There is a nominal clearance of 0.005 inches provided on the interior of the electromagnet supports to ensure a tight fit with the low-carbon steel rod.

Table 6.2: Engineering standards.

6.3 Evaluation of Design Tables

RQ1: Low heat generation

RQ2: Stable permanent magnet configuration

RQ3: Stable side electromagnets

RQ4: Individual control of the track segment electromagnets

RQ5: VL53L0X sensors used to detect the train's position

RQ6: Scalable circuit design

RQ7: Scalable train and track design

Design 1: Full track levitation (30 AWG)

Design 2: Track segment electromagnet control (20 AWG)

	Eval. of RQ1	Eval. of RQ2	Eval. of RQ3	Eval. of RQ4	Eval. of RQ5	Eval. of RQ6	Eval. of RQ7
Design 1	X	X	X	✓	✓	✓	✓
Design 2	✓	X	X	✓	✓	✓	✓

Table 6.3: Iterative process.
(✓ - Standards met, X - Standards not met)

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Appendix A

Building and Test Design Images



Figure A.1: Small permanent magnet halbach array.



Figure A.2: 1.25 inch permanent magnet strength test.



Figure A.3: 1.25 inch permanent magnet strength test with train.



Figure A.4: 1.25 inch permanent magnets on track segments.



Figure A.5: Illustration of aligned track magnets. The center magnet has the same outward-facing pole, yet it is still attracted toward the center of the track magnet configuration.



Figure A.6: X-shaped magnet configuration.

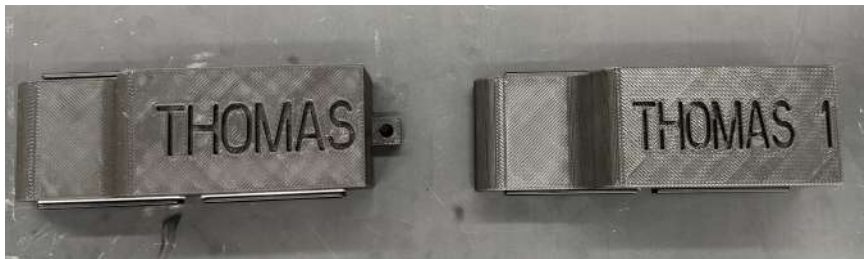


Figure A.7: Both iterations of the train design.



Figure A.8: Both iterations of the train design (bottom view).

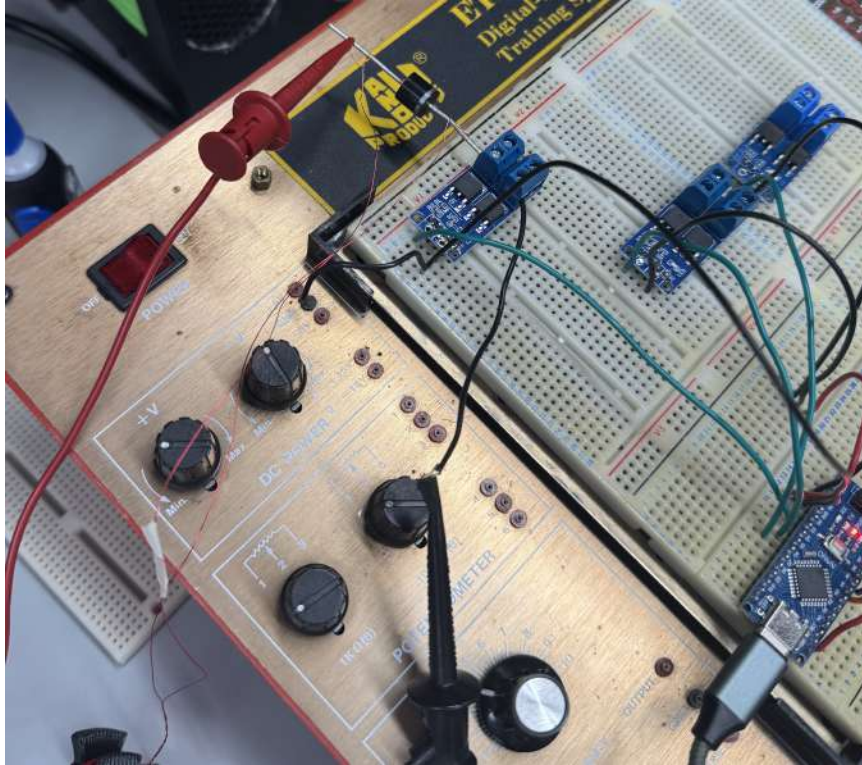


Figure A.9: Test of the electromagnet sub-circuit.



Figure A.10: Full track assembly.

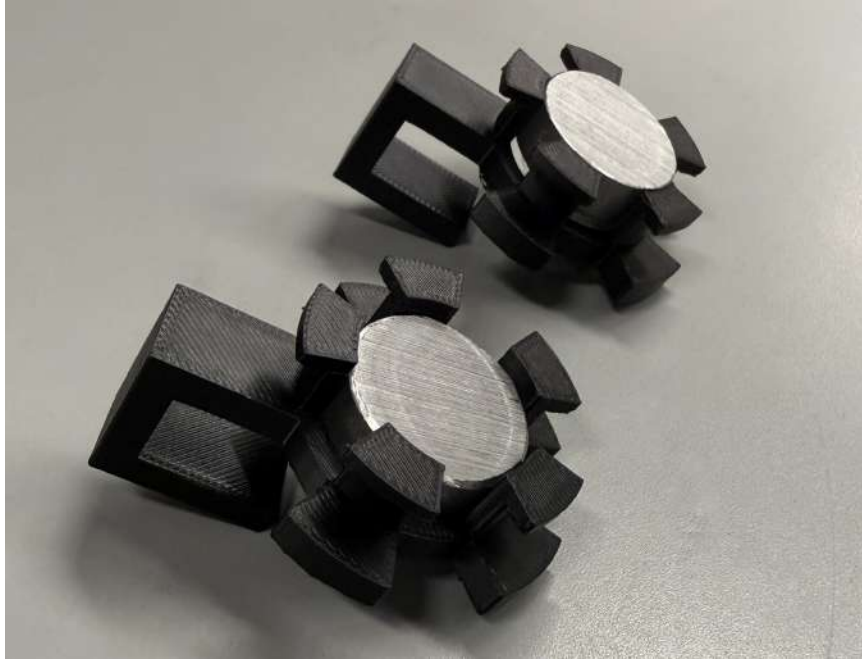


Figure A.11: Electromagnet holder with low-carbon steel core.



Figure A.12: Assembly of the final track design magnets. Each magnet was taped and glued to the bottom of the track.



Figure A.13: Assembly of the final track design electromagnets. Each electromagnet was wound around an electromagnet holder, and the Time-of-Flight (VL53L0x) sensors were screwed into broken holder pieces.

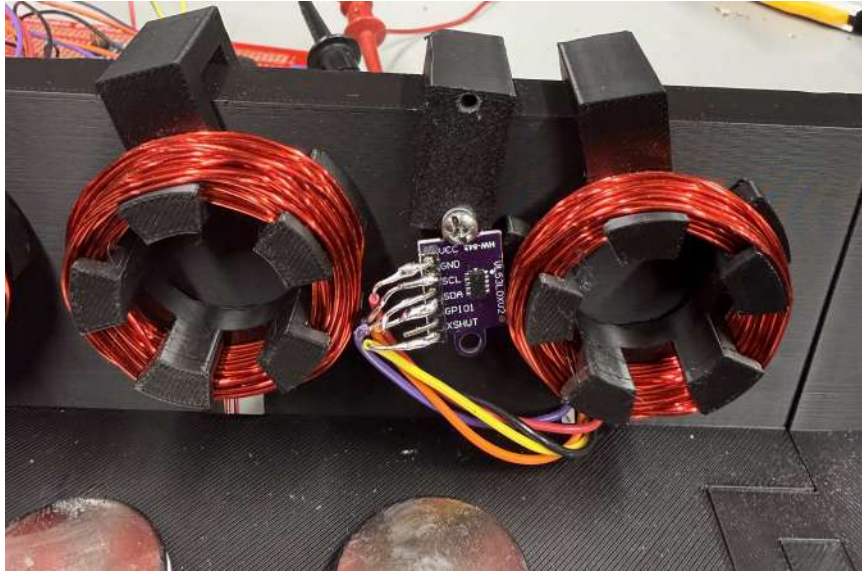


Figure A.14: Full assembly of the electromagnet and Time-of-Flight (VL53L0x) sensors.



Figure A.15: Top view of final track assembly.

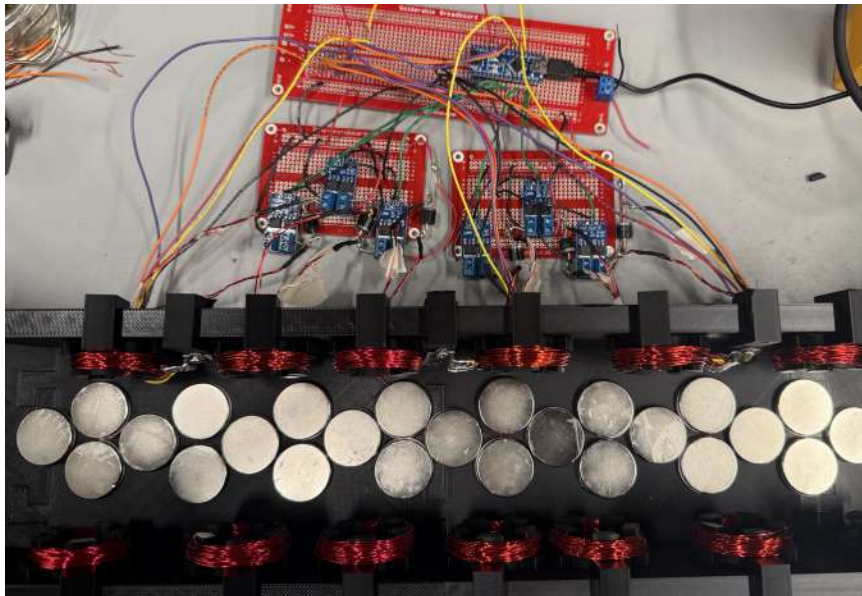


Figure A.16: Top view of electromagnet circuit final assembly.

Appendix B

Model Track Width and Clearance Code

This appendix includes the MATLAB code used to determine the minimum track width required for the train's motion.

```

%% Train-Track Clearance with COM on Centerline
% The train is a rectangle:
%   - Width W (radial) = 2 in
%   - Length L (tangential) = 6 in
% The train's center of mass (COM)
% moves along a circular centerline of radius r_c.
% The 2 in side is radial (from COM toward the track edges).
% Check that the train does not touch the track anywhere along the curve.

```

```

L = 6.0; % train length (inches)
W = 2.5; % train width (inches)

```

```

r_inner = 8.0; % inner track radius (inches)
clearance = 0.75; % radial clearance on each side (inches)

```

```

%% Derived track geometry

```

```

% Centerline radius of train COM (ensure clearance to inner rail)
r_c = r_inner + clearance + W/2;

```

```

% Outer rail radius providing same clearance on outer side
r_outer = r_c + W/2 + clearance;

```

```

track_width = r_outer - r_inner; % equals W + 2*clearance

```

```

fprintf('Train length L           = %.3f in\n', L);

```

```

Train length L           = 6.000 in

```

```

fprintf('Train width W           = %.3f in\n', W);

```

```

Train width W           = 2.500 in

```

```

fprintf('Side clearance           = %.3f in\n', clearance);

```

```

Side clearance           = 0.750 in

```

```

fprintf('Inner rail radius          = %.3f in\n', r_inner);

```

```

Inner rail radius        = 8.000 in

```

```

fprintf('Train center radius r_c = %.3f in\n', r_c);

```

```

Train center radius r_c = 10.000 in

```

```

fprintf('Outer rail radius          = %.3f in\n', r_outer);

```

```

Outer rail radius        = 12.000 in

```

```

fprintf('Track width                = %.3f in\n\n', track_width);

```

```

Track width              = 4.000 in

```

```

%% Sampling setup for clearance and fit check
% COM is constrained to radius r_c, train body is always tangent to centerline.

nTheta = 200; % samples along the track curve
nS      = 20;  % samples along train length
nY      = 20;  % samples across train width

% Angle range for the curved segment (example: 90 degree arc)
theta_vals = linspace(0, pi/2, nTheta);

% Local coordinates (s along length, y across width)
s_vals = linspace(-L/2, L/2, nS); % -L/2 (rear) to +L/2 (front)
y_vals = linspace(-W/2, W/2, nY); % -W/2 (inner edge) to +W/2 (outer edge)

%% Clearance evaluation (using radial projection, not full norm)
minRad      = inf;
maxRad      = -inf;
touch_inner = false;
touch_outer = false;
tol         = 1e-6;

for theta_c = theta_vals
    % Basis at train COM:
    ex = [-sin(theta_c); cos(theta_c)]; % tangent unit vector (direction of travel);
    % ex: 90 deg ahead of er
    er = [ cos(theta_c); sin(theta_c)]; % radial unit vector from origin to COM

    for s = s_vals
        for y = y_vals
            % Global position of a point (s, y) on the train
            P = (r_c + y) * er + s * ex;

            % Radial distance = projection onto er (NOT full norm)
            r_point = dot(P, er);

            minRad = min(minRad, r_point);
            maxRad = max(maxRad, r_point);

            % Inner: violation only if closer than (r_inner + clearance)
            if r_point < r_inner + clearance - tol
                touch_inner = true;
            end

            % Outer: violation only if farther than (r_outer - clearance)
            if r_point > r_outer - clearance + tol
                touch_outer = true;
            end
        end
    end
end
end
end

```

%% Results

```
fprintf('Minimum radial position of train: %.3f in\n', minRad);
```

Minimum radial position of train: 8.750 in

```
fprintf('Maximum radial position of train: %.3f in\n', maxRad);
```

Maximum radial position of train: 11.250 in

```
fprintf('Allowed radial range (no contact): (%.3f, %.3f) in\n', ...  
        r_inner + clearance, r_outer - clearance);
```

Allowed radial range (no contact): (8.750, 11.250) in

```
if touch_inner  
    fprintf(['\nInner side: TRAIN VIOLATES ' ...  
            'the requested clearance (too close to inner rail).\n']);  
else  
    fprintf(['\nInner side: Train maintains ' ...  
            'the requested clearance to inner rail.\n']);  
end
```

Inner side: Train maintains the requested clearance to inner rail.

```
if touch_outer  
    fprintf(['Outer side: TRAIN VIOLATES ' ...  
            'the requested clearance (too close to outer rail).\n']);  
else  
    fprintf(['Outer side: Train maintains ' ...  
            'the requested clearance to outer rail.\n']);  
end
```

Outer side: Train maintains the requested clearance to outer rail.

```
if ~(touch_inner || touch_outer)  
    fprintf(['\nOverall: Train length and width fit within ' ...  
            'the inner and outer track WITHOUT touching.\n']);  
else  
    fprintf(['\nOverall: Geometry does NOT satisfy the requested ' ...  
            'clearance somewhere along the curve.\n']);  
end
```

Overall: Train length and width fit within the inner and outer track WITHOUT touching.

Appendix C

Arduino Nano Propulsion Code

```
#include "Wire.h"
#include "VL53L0X.h"
```

```
// — Electromagnet pins (right → left)
```

```
const int emPin[6] = {11, 10, 9, 6, 5, 3};
```

```
// — XSHUT pins
```

```
—
const int XSHUT_C = A1;
const int XSHUT_B = A2;
const int XSHUT_A = A0;
```

```
// — Duty cycles
```

```
const int DUTY_OFF = 0;
const int DUTY_50 = 128;
const int DUTY_90 = 230;
const int DUTY_100 = 255;
```

```
// — Ramp settings
```

```
const unsigned long RAMP_STEP_MS = 5;
const int          RAMP_STEP_SIZE = 3;
```

```
// — Current and target duty cycle per magnet
```

```
int          currentDuty[6] = {0, 0, 0, 0, 0, 0};
int          targetDuty[6] = {0, 0, 0, 0, 0, 0};
unsigned long lastRamp[6]   = {0, 0, 0, 0, 0, 0};
```

```
// — Sensor objects and addresses
```

```
VL53L0X      sensorA, sensorB, sensorC;
const uint8_t ADDR_A = 0x30;
const uint8_t ADDR_B = 0x31;
const uint8_t ADDR_C = 0x32;
```

```
// — Detection threshold
```

```
const int DETECT_MM = 70;
```



```
// — Serial print throttle
```

```
const unsigned long printInterval = 100;
unsigned long      lastPrintTime = 0;
```

```
// — Sensor layout reference
```

```
// RIGHT
```

```
LEFT
```

```
// D11 [SensorC/A1] D10 D9 [SensorB/A2] D6 D5 [SensorA/A0] D3
// idx0          idx1 idx2          idx3 idx4          idx5
```

```
//
```

```
void testPins() {
  Serial.println("Starting pin test — each pin will fire at 50% for 500ms...");
  for (int i = 0; i < 6; i++) {
    Serial.print("Testing pin D");
    Serial.print(emPin[i]);
    Serial.println("...");
    analogWrite(emPin[i], DUTY_50);
    delay(500);
    analogWrite(emPin[i], DUTY_OFF);
    delay(200);
  }
  Serial.println("Pin test complete");
}
```

```
//
```

```
void initSensors() {
  digitalWrite(XSHUT_A, LOW);
  digitalWrite(XSHUT_B, LOW);
  digitalWrite(XSHUT_C, LOW);
  delay(50);

  digitalWrite(XSHUT_A, HIGH);
  delay(50);
  Serial.println("Attempting SensorA init...");
  if (!sensorA.init()) { Serial.println("ERROR: SensorA failed — check A0 wiring"); while (1); }
  sensorA.setAddress(ADDR_A);
```

```
sensorA.setTimeout(500);
sensorA.setMeasurementTimingBudget(20000); // 20ms — fast enough for responsive control
Serial.println("SensorA OK at 0x30 (leftmost, A0)");
```

```
digitalWrite(XSHUT_B, HIGH);
delay(50);
Serial.println("Attempting SensorB init...");
if (!sensorB.init()) { Serial.println("ERROR: SensorB failed — check A2 wiring"); while (1); }
sensorB.setAddress(ADDR_B);
sensorB.setTimeout(500);
sensorB.setMeasurementTimingBudget(20000);
Serial.println("SensorB OK at 0x31 (middle, A2)");
```

```
digitalWrite(XSHUT_C, HIGH);
delay(50);
Serial.println("Attempting SensorC init...");
if (!sensorC.init()) { Serial.println("ERROR: SensorC failed — check A1 wiring"); while (1); }
sensorC.setAddress(ADDR_C);
sensorC.setTimeout(500);
sensorC.setMeasurementTimingBudget(20000);
Serial.println("SensorC OK at 0x32 (rightmost, A1)");
```

```
Serial.println("All sensors ready");
```

```
}
```

```
//
```

```
void updateRamps() {
  unsigned long now = millis();
  for (int i = 0; i < 6; i++) {
    if (currentDuty[i] == targetDuty[i]) continue;

    if (now - lastRamp[i] >= RAMP_STEP_MS) {
      lastRamp[i] = now;

      if (currentDuty[i] < targetDuty[i]) {
        currentDuty[i] = min(currentDuty[i] + RAMP_STEP_SIZE, targetDuty[i]);
      } else {
        currentDuty[i] = targetDuty[i];
      }
    }

    analogWrite(emPin[i], currentDuty[i]);
  }
}
```

```
}  
}
```

```
//
```

```
int readDistance(VL53L0X &s) {  
    int d = s.readRangeSingleMillimeters();  
    if (s.timeoutOccurred() || d >= 8190) return -1; // Filter out-of-range reads  
    return d;  
}
```

```
//
```

```
void setup() {  
    for (int i = 0; i < 6; i++) {  
        pinMode(emPin[i], OUTPUT);  
        digitalWrite(emPin[i], LOW);  
    }
```

```
    pinMode(XSHUT_A, OUTPUT);  
    pinMode(XSHUT_B, OUTPUT);  
    pinMode(XSHUT_C, OUTPUT);
```

```
    Serial.begin(9600);  
    delay(100); // Allow Serial to stabilize before first print  
    Wire.begin();
```

```
    testPins();  
    initSensors();
```

```
    // — Start position: SensorB (middle) —————  
    targetDuty[2] = DUTY_90; // D9 — adjacent to SensorB  
    targetDuty[3] = DUTY_90; // D6 — adjacent to SensorB  
    targetDuty[1] = DUTY_50; // D10 — one step out from SensorB  
    targetDuty[4] = DUTY_50; // D5 — one step out from SensorB  
}
```

```
//
```

```
void loop() {
```

// — Read raw distances

```
int distA = readDistance(sensorA);  
int distB = readDistance(sensorB);  
int distC = readDistance(sensorC);
```

```
bool detA = (distA != -1 && distA < DETECT_MM);  
bool detB = (distB != -1 && distB < DETECT_MM);  
bool detC = (distC != -1 && distC < DETECT_MM);
```

// — Track whether anything changed this loop

```
bool stateChanged = false;  
char cmd = 0;
```

// — Key input

```
if (Serial.available() > 0) {  
    cmd = Serial.read();  
    stateChanged = true;  
}
```

// — Only recompute targets if a sensor fired or key was pressed

```
if (detA || detB || detC || stateChanged) {  
    int t[6] = {DUTY_OFF, DUTY_OFF, DUTY_OFF, DUTY_OFF, DUTY_OFF, DUTY_OFF};
```

```
    if (detA) {  
        t[4] = max(t[4], DUTY_90); // D5 — adjacent to SensorA  
        t[5] = max(t[5], DUTY_90); // D3 — adjacent to SensorA  
        t[3] = max(t[3], DUTY_50); // D6 — one step out from SensorA  
    }
```

```
    if (detB) {  
        t[2] = max(t[2], DUTY_90); // D9 — adjacent to SensorB  
        t[3] = max(t[3], DUTY_90); // D6 — adjacent to SensorB  
        t[1] = max(t[1], DUTY_50); // D10 — one step out from SensorB  
        t[4] = max(t[4], DUTY_50); // D5 — one step out from SensorB  
    }
```

```
    if (detC) {  
        t[0] = max(t[0], DUTY_90); // D11 — adjacent to SensorC  
        t[1] = max(t[1], DUTY_90); // D10 — adjacent to SensorC  
        t[2] = max(t[2], DUTY_50); // D9 — one step out from SensorC  
        t[3] = max(t[3], DUTY_50); // D6 — one step out from SensorC  
    }
```

```
    if (cmd == 'r' || cmd == 'R') {
```

```

    if (detC) { Serial.println("Train at rightmost position"); }
    else if (detB) { t[1] = DUTY_100; Serial.println("Moving right → D10 at 100%"); }
    else if (detA) { t[3] = DUTY_100; Serial.println("Moving right → D6 at 100%"); }
    else { Serial.println("No train detected"); }
}

if (cmd == 'I' || cmd == 'L') {
    if (detA) { Serial.println("Train at leftmost position"); }
    else if (detB) { t[3] = DUTY_100; Serial.println("Moving left → D6 at 100%"); }
    else if (detC) { t[2] = DUTY_100; Serial.println("Moving left → D9 at 100%"); }
    else { Serial.println("No train detected"); }
}

for (int i = 0; i < 6; i++) {
    targetDuty[i] = t[i];
}
}

updateRamps();

// — Serial output (rate limited) —————
unsigned long now = millis();
if (now - lastPrintTime >= printInterval) {
    lastPrintTime = now;

    Serial.print("A: ");
    Serial.print(distA == -1 ? "TIMEOUT" : String(distA) + "mm");
    Serial.print(detA ? " [A ACTIVE]" : "");

    Serial.print(" | B: ");
    Serial.print(distB == -1 ? "TIMEOUT" : String(distB) + "mm");
    Serial.print(detB ? " [B ACTIVE]" : "");

    Serial.print(" | C: ");
    Serial.print(distC == -1 ? "TIMEOUT" : String(distC) + "mm");
    Serial.print(detC ? " [C ACTIVE]" : "");

    Serial.println();
}
}

```

Appendix D

Arduino Nano Track Segment ToF Code

```

#include "Wire.h"
#include "VL53L0X.h"

VL53L0X sensor;

const int mosfetPin1 = 3; // PWM pin - 200 to 400mm
const int mosfetPin2 = 5; // PWM pin - 400 to 1000mm
const int mosfetPin3 = 6; // PWM pin - 1000 to 1500mm
const int duty10 = 75;    // ~10% of 255  $\approx$  25.5  $\rightarrow$  26

const int near_low  = 50; // Pin 3 range
const int near_high = 120;
const int mid_low   = 120; // Pin 5 range
const int mid_high  = 200;
const int far_low   = 200; // Pin 6 range
const int far_high  = 500;

const unsigned long printInterval = 100;
unsigned long lastPrintTime = 0;

void sensor_init(bool long_range, bool high_speed) {
  Wire.begin();
  sensor.setTimeout(500);
  sensor.init();
  if (long_range) {
    sensor.setSignalRateLimit(0.1);
    sensor.setVcselPulsePeriod(VL53L0X::VcselPeriodPreRange, 18);
    sensor.setVcselPulsePeriod(VL53L0X::VcselPeriodFinalRange, 14);
  }
  int budget = high_speed ? 20000 : 200000;
  sensor.setMeasurementTimingBudget(budget);
}

int get_distance() {
  return sensor.readRangeSingleMillimeters();
}

void setup() {
  pinMode(mosfetPin1, OUTPUT);
  pinMode(mosfetPin2, OUTPUT);
  pinMode(mosfetPin3, OUTPUT);
  Serial.begin(9600);
  sensor_init(false, false);
}

```

```

void loop() {
  int distance = get_distance();

  // Guard against bad/timeout readings
  if (sensor.timeoutOccurred()) {
    Serial.println("Sensor timeout");
    analogWrite(mosfetPin1, 0);
    analogWrite(mosfetPin2, 0);
    analogWrite(mosfetPin3, 0);
    return;
  }

  bool pin1Active = (distance >= near_low && distance <= near_high);
  bool pin2Active = (distance >= mid_low && distance <= mid_high);
  bool pin3Active = (distance >= far_low && distance <= far_high);

  // Pin 3 - 200 to 400mm inclusive
  analogWrite(mosfetPin1, pin1Active ? duty10 : 0);

  // Pin 5 - 400 to 1000mm inclusive
  analogWrite(mosfetPin2, pin2Active ? duty10 : 0);

  // Pin 6 - 1000 to 1500mm inclusive
  analogWrite(mosfetPin3, pin3Active ? duty10 : 0);

  // Serial printing rate-limited to avoid slowing the loop
  unsigned long now = millis();
  if (now - lastPrintTime >= printInterval) {
    lastPrintTime = now;

    if (pin1Active) {
      Serial.print("Distance: ");
      Serial.print(distance);
      Serial.println(" mm | Pin3: ON");
    } else if (pin2Active) {
      Serial.print("Distance: ");
      Serial.print(distance);
      Serial.println(" mm | Pin5: ON");
    } else if (pin3Active) {
      Serial.print("Distance: ");
      Serial.print(distance);
      Serial.println(" mm | Pin6: ON");
    } else {

```



```
    Serial.print("Distance: ");  
    Serial.print(distance);  
    Serial.println(" mm | No pin activated: OFF");  
  }  
}  
}
```